Software Development 2

BSCH-SD2

TDD Project

Date

GRIFFITH COLLEGE DUBLIN

Table of Contents

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Faculty:	Computing Science					
Course:	BSCH/BSCO/EXCH			Stage/year:	2	
Subject:	Software Development 2					
Study Mode:	Full time	Yes		Part-time	No	
Lecturer Name:	Tracey Cassells					
Assignment Title:	SD2 Project - Final Documentation					
Date due:	Monday, 27 April 2026, 11:59 PM					
Date submitted:	Monday,27 April 2026, 2pm					

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I hereby certify that this assignment is my own work, based on my personal study and/or research, and that I have acknowledged all material and sources used in its preparation. I also certify that the assignment has not previously been submitted for assessment and that I have not copied in part or whole or otherwise plagiarised the work of anyone else, including other students.

Signed: Yun Long Wang **Date:** 2026/4/27
Signed: Yu tian Wang **Date:** 2026/4/27
Signed: Jing zu Ge **Date:** 2026/4/27
Signed: **Date:**

1. Overview

1.1 Goals

//Aims of the project

//Explanation of game rules

//Features your game has

2. Development Process

//The steps you took to bring the project from idea to finished product

2.1 Research

//New technologies researched for the project, e.g. Swing Javax

3. External Packages

//Explanation of any external libraries/ APIs used

4. Project Setup

//The folder layout, instructions to run your project, include a Class Diagram

5. Milestone 1

5.1 Goals

Milestone aims: I want to do the most basic interface well (for yunlong wang). Calculate the most basic algorithms well. The main purpose is to develop the prototype of the game we want to play, Gomoku.

Achievements: Yunlong Wang: Mainly responsible forbuilding windows andcreating several basic buttons (such as start game and help buttons, etc.). But this time we did it purely individually, without integrating each

other's assignments.

Jingzu Ge: In Milestone 1, he managed the chessboard, judged the outcome of the game, obtained pieces,

determined if the chessboard was full, and reset the chessboard. But these algorithms have not been integrated.

In milestone 2, we will merge together.

Yutian Wang: Responsible for initializing the GitLab repository and building the project skeleton.

Implement turn based state management and develop backend logic for Timekeeping and Timeout loss.

Issues: Bugs. Because this is what we just started to do, we didn't have a clue at the beginning. Later, we gradually learned what to do through the teaching video on the website and consulting other students.

5.2 Testing

Explanation of what tests were added and why.

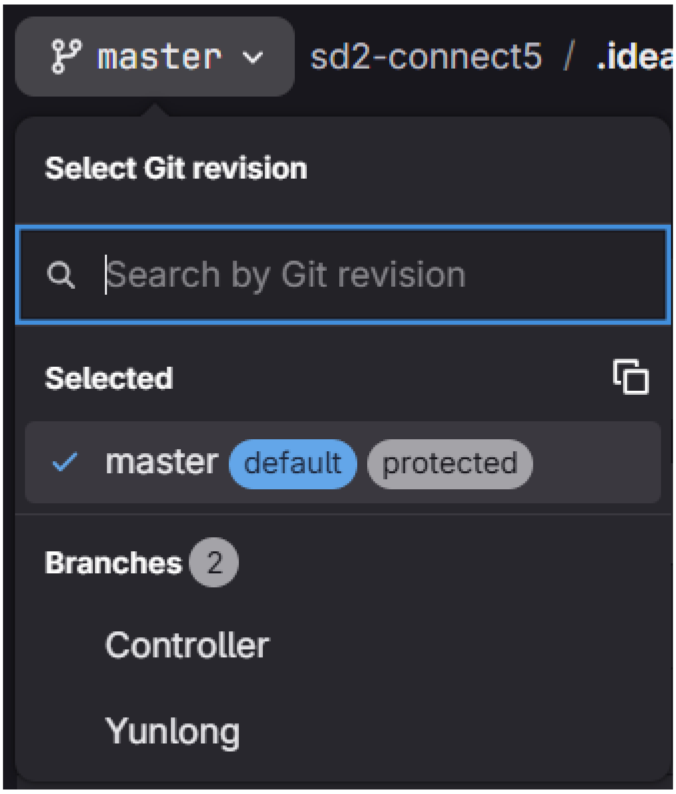
Answer: This time, only Yunlong Wang's homework added the test file. I'm sorry that Jingzu Ge and Yutian Wang's homework did not include the test file. Yunlong Wang's test mainly involves running windows to display the basic graphics of the game.

5.3 Commit Logs & Branches Tree

Commit Logs:

Displayed on Gitlab (granting developer rights to teachers)

Branches Tree



Yunlong Wang did :Yunlong this branch.

YutianWangdid:Controllerthisbranch.

Jingzu Ge did:master this branch.

6. Milestone 2

6.1Goals

Milestone aims: First, we add the undo function, so player can take back last move when they make mistake. Second, we add countdown timer, each player have 5 minutes, if time out they will lose. Third, we add simple AI, AI will randomly put piece on empty position, this is good for beginner to practice. Fourth, we add merge mode, when player have three consecutive same color pieces, they can merge and capture enemy pieces around. Fifth, we add endgame mode, player can load preset board from txt file and try to win in limited steps. Sixth, we improve user interface, add time display, turn indicator and more buttons. Finally, I am responsible for combine everyone code together, because three people work separately, I need fix conflict and make sure everything work. The main challenge is merging different code style and make sure all functions can run without error. After testing, basic functions work well, but AI is still very simple, we may improve in the future. Overall, Milestone 2 achieve most aims we set.

Achievements: On the basis of milestone 1: Yunlong Wang improved the button design of AI mode and the selection option buttons of different difficulty (classic difficulty and merge difficulty). And Yunlong Wang linked the algorithm code of Jingzu Ge with his boardpanel, so that the program can run these modes by clicking the button. It also improves the repentance function. And it also realizes the important function that five chess matches will determine whether to win or lose. In a word, at this stage, what Yunlong Wang mainly did was to connect with the code of Jingzu Ge, so that many functions were realized, and corresponding function buttons were added.

Jingzu Ge: The original algorithm and logic have been modified, and the game switching mode and simple ai have been introduced. These two algorithms have been introduced, and an interface has been provided for functions such as repenting chess, resetting the board, and playing chess. The test class has been created independently to verify the accuracy and implementability of each code.By docking with the Cloud Dragon King, the code is effectively linked to realise the basic operation of the game.

Yutian Wang: write the core logic of "undo" (use the stack structure to record the historical steps). Realize the underlying rule determination of the special mechanism of "merging chess pieces". Build a level data loading system in the residual mode.

Issues: 1. Our AI model is too simple. We tried a harder AI model, but encountered a lot of errors. So we made the most basic non automatic AI mode (which means you can click the button all the time to run AI mode when you start the game). We will be more perfect in the future homework. 2. When it is difficult to make

merge, the code controlled by the chessboard is forgotten to be modified so that the merge cannot run normally.

At that time, we just blindly modified the algorithm file and ignored the modification of the chessboard interface.

3. The code is often repeated or misspelled when writing words, and many careless errors have occurred.4. And

another important point is that Yunlong Wang found that after organizing and integrating Jingzu Ge's code, she

actually integrated Yutian Wang's code separately. For some reason, our Long and Ge code cannot be adapted

to her code (if we insist on adapting, it will be difficult and make the code unsightly and complex). So later,

Yunlong Wang changed a method and asked Yutian Wang to continue working on her code with Ge and Long's

code as the main focus, and Yunlong Wang helped integrate her code.

6.2 Testing

Explanation of what tests were added and why.

Answer: Because Ge and Yutian did not write the test in the first stage of submission, they both completed

it in this stage

6.3 Commit Logs & Branches Tree

Commit Logs:

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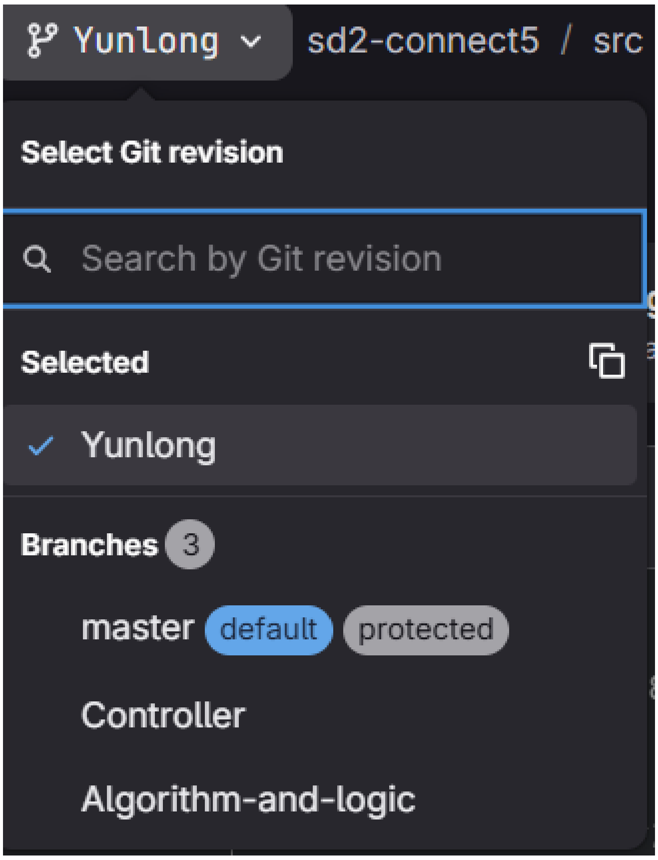
Branches Tree



Jingzu Ge did “Algorithm-and-logic” this branch.

Yutian Wang did “Controller” this branch.

Yunlong Wang did “Yunlong” this branch.

“master” this branch is out-dating:

7. Milestone 3

7.1 Goals

Milestone aims: Our goal at this stage. In fact, compared to the first and second stages, it is relatively easy at this stage. Because in the first stage, we had no clue and needed to acquire a lot of knowledge and ideas. In Stage 2, we had ideas and frameworks, but we only continued to complete them within the framework of Stage 1. So we are essentially building the framework of a house without modifying the interior of the house. Completing the second stage means that the house has already taken shape. So in the third stage, we just need to decorate the house to make it more complete and beautiful. So at this stage, we start adding more complete features to make the UI look better. Also, investigate some errors.

Achievements: Everyone in our group has a clear division of labor. The following is the exact division of labor for everyone: Yunlong Wang: in the process of milestone 3, he mainly optimized the UI to make it more like a truly complete game. And he integrated the code of the other two members completely. Yutian Wang: she mainly completed the code of merge mode in the process of milestone 3. Jingzu Ge: he mainly completed the AI mode in the process of milestone 3. And multiplayer mode.

Issues: We inevitably encountered some problems when we were doing group work. For example, when designing the UI, I (Yunlong Wang) wanted to install some pictures that could highlight functions into my UI interface. After writing the code, I directly ran my code, but it didn't run successfully in the end. Finally, I found that I didn't write the code to load pictures in my boardpanel.java file at first. And I had a lot of bugs during the integration (for example, there were a lot of code duplication in the code integration between me and the other two members, and some names of Gobang function codes were different, but I finally corrected them to make them run normally).

7.2 Testing

Explanation of what tests were added and why.

7.3 Commit Logs & Branches Tree

Commit Logs:

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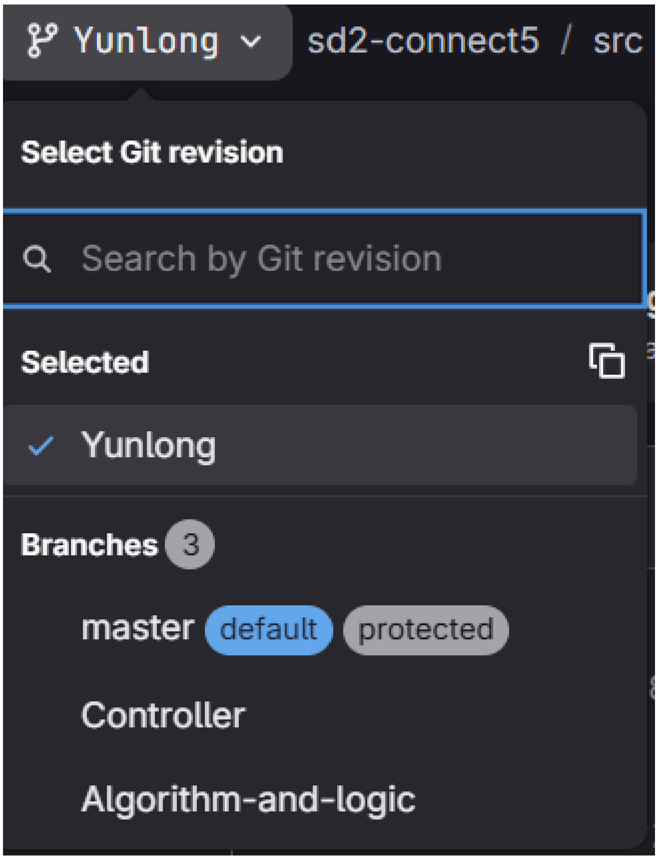

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8. Bibliography

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